

printed page 1104 equation (5.5)

$$\rho_{X_0,t_0}(F,t)=\frac{1}{4\pi(t_0-t)}\exp\left(-\frac{|F-X_0|^2}{4(t_0-t)}\right),$$

cutoff $\phi\in C_0^\infty(B_{2r}(X_0))$ $\phi\equiv 1$ on $B_r(X_0)$ localized Gaussian quantity

$$\Psi(X_0,t_0;t)=\int_{\Sigma_t}(\cos\alpha)^{-p}\phi(F)\rho_{X_0,t_0}(F,t)\,d\mu_t.$$

Theorem 3.3 differential inequality

$$\frac{d}{dt}\left(e^{c_1\sqrt{t_0-t}}\Psi(X_0,t_0;t)\right)\leq -\text{nonnegative square terms}+c_2e^{c_1\sqrt{t_0-t}}.$$

Section 5 parabolic rescaling

$$\begin{aligned}F_\lambda(x,s)&=\lambda\big(F(x,T+\lambda^{-2}s)-X_0\big),\qquad s<0,\\d\mu_s^\lambda&=\lambda^2d\mu_{T+\lambda^{-2}s},\qquad w_\lambda=(\cos\alpha_\lambda)^{-p}.\end{aligned}$$

printed equation (5.5) fixed $-1< s_1< s_2< 0$

$$G_\lambda(s_2)-G_\lambda(s_1)\rightarrow 0\quad\text{as }\lambda\rightarrow\infty,$$

endpoint quantity

$$G_\lambda(s)=e^{c_1\sqrt{T-(T+\lambda^{-2}s)}}\int_{\Sigma_s^\lambda}w_\lambda\phi_R(F_\lambda)(-s)^{-1}\exp\left(-\frac{|F_\lambda|^2}{4(-s)}\right)d\mu_s^\lambda.$$

gradient flow printed page 1104 equation (5.5)

exact gap corrected claim remaining moving-cutoff lemma

lemma false / underived strongest corrected theorem

Proven **cutoff**

$\chi\in C_0^\infty(B_{2r}(X_0))$ admissible cutoff $\chi=1$ on $B_r(X_0)$

$$\Psi_\chi(t)=\int_{\Sigma_t}(\cos\alpha)^{-p}\chi(F)\frac{1}{4\pi(T-t)}\exp\left(-\frac{|F-X_0|^2}{4(T-t)}\right)d\mu_t,$$

$$Q_\chi(t)=e^{c_1\sqrt{T-t}}\Psi_\chi(t).$$

differential inequality nonpositive square terms

$$Q_\chi'(t) \leq c_2 e^{c_1 \sqrt{T-t}}.$$

$$[a,T) \quad \text{integrable}$$

$$A(t)=\int_t^{t'}c_2e^{c_1\sqrt{T-u}}\,du.$$

$$A(t)\rightarrow 0$$

$$(Q_\chi+A)'(t)=Q_\chi'(t)-c_2e^{c_1\sqrt{T-t}}\leq 0.$$

$$Q_\chi+A \text{ monotone decreasing} \quad \text{bounded below} \quad Q_\chi(t) \quad \text{finite one-sided limit}$$

$$\lim_{t\uparrow T}Q_\chi(t)=L_\chi<\infty.$$

$$\text{exact rescaling}$$

$$t_\lambda(s)=T+\lambda^{-2}s.$$

$$T-t_\lambda(s)=\lambda^{-2}(-s),\qquad F-X_0=\lambda^{-1}F_\lambda,\qquad d\mu_s^\lambda=\lambda^2d\mu_{t_\lambda(s)}.$$

$$\text{K\"ahler angle} \quad \text{positive homothety} \quad \text{invariant}$$

$$(\cos\alpha_\lambda)^{-p}=(\cos\alpha)^{-p}.$$

$$\begin{aligned} &\int_{\Sigma_s^\lambda}(\cos\alpha_\lambda)^{-p}\chi(X_0+\lambda^{-1}F_\lambda)(-s)^{-1}\exp\left(-\frac{|F_\lambda|^2}{4(-s)}\right)d\mu_s^\lambda\\ &=4\pi\,\Psi_\chi(t_\lambda(s)). \end{aligned}$$

$$\text{exponential factor}$$

$$J_\lambda^\chi(s)=4\pi Q_\chi(t_\lambda(s)).$$

$$\text{fixed } s_1 < s_2 < 0$$

$$J_\lambda^\chi(s_2)-J_\lambda^\chi(s_1)=4\pi\big(Q_\chi(t_\lambda(s_2))-Q_\chi(t_\lambda(s_1))\big)\rightarrow 0,$$

$$\text{terminal limit } L_\chi$$

$$\begin{array}{ccccc} 4\pi & \text{normalization} & \text{normalized kernel } [4\pi(-s)]^{-1} & \text{factor} & \text{unnormalized} \\ (-s)^{-1} & 4\pi & \text{zero limit} & & \end{array}$$

$$\textbf{Proven} \quad \textbf{printed fixed scaled cutoff} \quad \textbf{fixed original cutoff}$$

$$\text{printed expression} \qquad \phi_R(F_\lambda) \qquad \phi_R \text{ fixed in scaled coordinates} \qquad \text{original coordinates}$$

$$\phi_R(F_\lambda)=\phi_R(\lambda(F-X_0))=: \chi_{R,\lambda}(F).$$

$$\chi_{R,\lambda}(X)=\phi_R(\lambda(X-X_0)).$$

$$\text{cutoff} \qquad B_{2R/\lambda}(X_0) \qquad B_{R/\lambda}(X_0) \qquad 1 \qquad \lambda$$

$$\text{printed endpoint quantity}$$

$$4\pi$$

$$\begin{array}{llllll} \text{Section 3 type quantity} & \text{quantity} & \text{cutoff} & \chi_{R,\lambda} & \text{fixed } \chi & Q_\chi(t) \\ \text{terminal-limit argument} & \text{moving family} & Q_{\chi_{R,\lambda}}(t) & & & \end{array}$$

$$\textbf{Proven} \quad \textbf{individual terminal limits for moving families} \qquad \textbf{two-time limit}$$

$$Q_n(t)=n^2(-t), \qquad t\in (-1,0).$$

$$Q_n\geq 0 \quad \text{locally absolutely continuous}$$

$$Q'_n(t)=-n^2\leq 0.$$

$$Q_n \qquad \text{terminal limit}$$

$$\lim_{t\uparrow 0}Q_n(t)=0.$$

$$\text{fixed } s_1 < s_2 < 0$$

$$Q_n(n^{-2}s_2)-Q_n(n^{-2}s_1)=s_1-s_2\neq 0.$$

$$\text{“} \qquad \text{moving cutoff quantity} \qquad \text{terminal limit”} \qquad \text{imply diagonal parabolic-scale two-time convergence}$$

$$\textbf{Proven} \quad \textbf{literal fixed scaled compact cutoff} \quad \textbf{static plane model}$$

$$\text{flat symplectic plane } P=\mathbb{R}^2\times\{0\}\subset\mathbb{C}^2 \qquad \text{stationary immersion } F(x,t)=x$$

$$\cos\alpha=1, \qquad H=0, \qquad V=0,$$

$$\text{parabolic rescaling} \qquad \text{plane} \qquad \text{weight} \quad 1$$

$$\text{nontrivial radial compact cutoff } \phi_R$$

$$\phi_R=1 \text{ on } B_R, \qquad \phi_R=0 \text{ outside } B_{2R},$$

$$\text{radial nonincreasing}$$

$$I(s)=\int_P\phi_R(Y)(-s)^{-1}\exp\left(-\frac{|Y|^2}{4(-s)}\right)dA(Y).$$

$$\text{uncut integral}$$

$$\int_P (-s)^{-1} \exp\left(-\frac{|Y|^2}{4(-s)}\right) dA(Y) = 4\pi.$$

compactly cut integral $s_1 = -1 \quad s_2 = -1/4 \quad Y = \sqrt{a}Z \quad a = -s$

$$I(-a) = \int_P \phi_R(\sqrt{a} \, Z) e^{-|Z|^2/4} dA(Z).$$

$a_2 = 1/4 < a_1 = 1 \quad \phi_R$ radial nonincreasing

$$\phi_R(\sqrt{a_2}Z) \geq \phi_R(\sqrt{a_1}Z),$$

cutoff annulus

$$I(-1/4) > I(-1).$$

static plane model

$$G_\lambda(s) = e^{c_1 \lambda^{-1} \sqrt{-s}} I(s),$$

$$\lim_{\lambda \rightarrow \infty} \big(G_\lambda(-1/4) - G_\lambda(-1)\big) = I(-1/4) - I(-1) > 0.$$

disproves standalone fixed-scaled-cutoff lemma

plane full Section 5 compact first-singular-flow counterexample noncompact finite first
singular time $X_0 \quad T$ singular point

Proven moving cutoff derivative terms order-one annular flux

literal cutoff

$$\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0)),$$

$$D\chi_{R,\lambda} = \lambda D\phi_R, \qquad D^2\chi_{R,\lambda} = \lambda^2 D^2\phi_R.$$

pre-(5.5) monotonicity calculation cutoff-gradient cutoff-Laplacian heat-kernel-gradient terms
parabolic change of variables $t = T + \lambda^{-2}s$ fixed scaled annulus

$$R < |F_\lambda| < 2R$$

spacetime integrals

$$\int_{s_1}^{s_2} \int_{\Sigma_s^\lambda} w_\lambda \rho_s(F_\lambda) \left[D\phi_R(F_\lambda) \cdot v_\lambda + \Delta_{\Sigma_s^\lambda}(\phi_R \circ F_\lambda) + 2 \langle \nabla_{\Sigma_s^\lambda} \log \rho_s, \nabla_{\Sigma_s^\lambda}(\phi_R \circ F_\lambda) \rangle \right] d\mu_s^\lambda ds.$$

$\lambda^{-1} \quad \lambda^{-2}$ prefactor moving cutoff error scaling $o(1)$ literal
moving-cutoff endpoint limit annular flux vanishing / cancellation uniform moving-

cutoff convergence-to-terminal-limit

Conditional endpoint weighted measures nonzero cone/plane literal compact cutoff
positive defect

s_1, s_2 endpoint pushed-forward weighted measures nonzero two-dimensional cone
measure ν_{cone} suitable radial compact cutoff

$$\int \phi_R(Y)(-s_2)^{-1} e^{-|Y|^2/[4(-s_2)]} d\nu_{\text{cone}} > \int \phi_R(Y)(-s_1)^{-1} e^{-|Y|^2/[4(-s_1)]} d\nu_{\text{cone}}.$$

literal compact scaled cutoff endpoint difference limit strictly positive zero
conditional obstruction full counterexample same-limit conical convergence (5.5)
noncircularly

printed $\phi_R(F_\lambda)$ fixed in scaled coordinates original coordinates moving cutoff
 $\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0)).$

Theorem 3.3 terminal-limit argument fixed localized Gaussian quantity

$$Q_\chi(t) \rightarrow L_\chi,$$

moving family $Q_{\chi_{R,\lambda}}(t)$ parabolic-scale times uniformly terminal limit
moving cutoff monotonicity calculation cutoff derivative terms scaled
variables order-one annular flux small parameter sign static plane computation
flux velocity curvature flow-energy compact cutoff Gaussian width s
endpoint difference
fixed-cutoff monotonicity one-sided limit local area bound fixed-time Radon compactness
literal (5.5) noncircular bridge

stage	question addressed	conclusion established	effect on the approach
fixed-cutoff terminal limit	(3.6) one-sided inequality terminal limit	fixed cutoff χ $Q_\chi(t)$ has finite one-sided limit as $t \uparrow T$	fixed-cutoff endpoint route is valid
exact rescaling	rescaled endpoint quantity Section 3 quantity	unnormalized scaled integral equals $4\pi\Psi_\chi$ for fixed pullback cutoff $\chi(X_0 + \lambda^{-1}F_\lambda)$	4π is harmless normalization

stage	question addressed	conclusion established	effect on the approach
cutoff comparison	printed $\phi_R(F_\lambda)$ fixed-cutoff pullback	$\phi_R(F_\lambda)$ corresponds to moving cutoff $\chi_{R,\lambda}(X) = \phi_R(\lambda(X - X_0))$	identifies the real gap
formal moving-family test	individual terminal limits imply diagonal two-time convergence	false, even for monotone nonnegative scalar families	fixed-cutoff argument cannot be applied familywise
static plane model	fixed scaled compact cutoff intrinsically harmless	false in standalone model; endpoint difference can be strictly positive	shows the obstruction is real
annular flux audit	pre-(5.5) estimates hide a uniform moving-cutoff estimate	no; derivative terms become order-one annular flux	literal moving-cutoff lemma remains unproved
compactness audit	fixed-time local area compactness repairs the gap	only gives separate fixed-time weak limits, not same-limit or flux control	insufficient for literal endpoint limit
corrected theorem	what unconditional statement is actually proved	fixed-original-cutoff endpoint theorem with $\chi(X_0 + \lambda^{-1}F_\lambda)$	valid replacement for rigorous use

literal fixed compact scaled cutoff reading verified proof full compact first-singular-flow counterexample proved stronger diagnosis (3.6) printed (5.5) common-limit step fixed-original-cutoff replacement literal $\phi_R(F_\lambda)$ noncircular moving-cutoff theorem

unconditional corrected statement $\boxed{\begin{minipage}{0.92\linewidth} \text{Let } (X_0) \text{ be the blow-up point and } (T) \text{ the first singular time. Let } (\chi \in C_0^\infty(B_{2r}(X_0))) \text{ be fixed with } (\chi=1) \text{ on } (B_r(X_0)). \text{ For fixed } (-1 < s_1 < s_2 < 0), \text{ define } \text{@@MATH0@@} \text{ Then } \text{@@MATH1@@} \text{ The same statement with normalized kernel } ([-4\pi(-s)]^{-1}) \text{ is equivalent. } \end{minipage}} \}$

literal moving-cutoff lemma $\boxed{\begin{minipage}{0.92\linewidth} \text{For the moving cutoff } \text{@@MATH2@@} \text{ let } \text{@@MATH3@@} \text{ If } (L_{\{R,\lambda\}} = \lim_{\{t \uparrow T\}} Q_{\{R,\lambda\}}(t)), \text{ prove noncircularly that for each } (i=1,2), \text{ } \text{@@MATH4@@} \text{ Equivalently, prove annular cutoff-flux cancellation/vanishing on } \text{@@MATH5@@} \text{ at the parabolic scale } (T-t \sim \lambda^{-2}). \end{minipage}} \}$

bridge proved full-hypothesis counterexample constructed printed literal compact-
scaled-cutoff equation (5.5)